

1. (5%) In a data center, what is the appropriate definition of good performance? It (a) get a program done first (b) has the fast response time (c) completes the most jobs during a certain periods.

A: C.

2. (10%) By Amdahl's Law, a new method which can contribute 80% improvement on the 45% part of the entire system. What is the speedup? (speedup 可用%呈現)

$$F = 45\% \quad S = 1 + 80\% = 1.8 \quad \text{SPD UP} = \frac{1}{(1 - 0.45) + \frac{0.45}{1.8}} = \frac{1}{0.8} = 1.25 \#$$

3. (10%) Suppose we have two implementations of the same instruction set architecture. Computer A has a clock cycle time of 1500 ps, and computer B has a clock rate of 3.0 GHz. Consider a program having 1000 instructions.

(a) Suppose computer A has a CPI of 3.2 for the program. Find the CPU time (in ns) for the computer A.

(b) Suppose the CPU time of the computer B is 1000 ns for the same program. Compute the CPI of computer B for the program.

(a) $1000 \times 3.2 \times 1500 \times 10^{-12} = 4.8 \times 10^{-6}$ A: $4.8 \mu s$

(b) $\text{CPU Time} = \frac{\text{CPI} \times \text{Instruction}}{\text{Clock Rate}} \quad 1000 \times 10^{-9} = \frac{\text{CPI} \times 1000}{3.0 \times 10^9} \quad \text{CPI} = 3 \quad A: 3$

4. (10%) Case 1, Machine M with compiler X

Instruction class	CPI	Instruction count
A	3	2000
B	8	3000
C	6	1200
D	6	1500

Case 2, Machine M with compiler Y

Instruction class	CPI	Instruction count
A	3	1800
B	8	2500
C	5	1400
D	7	1800

- (a) Which case has better effects?

(a) Case 1: $3 \times 2000 + 8 \times 3000 + 6 \times 1200 + 6 \times 1500$
 $= 6000 + 24000 + 7200 + 9000$
 $= 46200$

Case 2: $3 \times 1800 + 8 \times 2500 + 5 \times 1400 + 7 \times 1800$
 $= 5400 + 20000 + 7000 + 12600$
 $= 45000 \quad (\checkmark)$

A: Case 2 is better

(b) How slow the clock rate should be to make the process has 0.4s difference?

$$46200 - 45000 = 1200 \text{ (clock cycles)}$$

$$1200 = \text{clock Rate} \times 0.4$$

$$\text{Clock Rate} = 3000$$

$$A: 3000 \text{ Hz}$$

5. (15%) The following table shows the number of **instructions** for a program. Assuming that the **CPI** for the **R-format**, **load**, **store**, and **branch** instructions is **1**, **5**, **4** and **3**, respectively.

¹ R-Format	⁵ Load	⁴ Store	³ Branch
70×10^9	25×10^9	15×10^9	30×10^9

70
125
60
90
345

- (a) Find the average CPI for the program.

$$(a) \quad 70 + 25 + 15 + 30 = 140$$

$$1 \times \frac{70}{140} + 5 \times \frac{25}{140} + 4 \times \frac{15}{140} + 3 \times \frac{30}{140} = \frac{345}{140} = 2.46 \#$$

- (b) By how much we can improve the CPI of Load instructions if we want the program to run 1.25 times faster?

★ 使用 instruction 代入 阿姆達爾

$$(b) \quad F = \frac{5 \times 25}{70 + 5 \times 25 + 4 \times 15 + 3 \times 30} = \frac{125}{345}$$

$$1.25 = \frac{1}{\frac{220}{345} + \frac{125/5}{345}} \quad \frac{345}{220 + 125/5} = 1.25 \quad S = 2.23 \#$$

$$CPI_{\text{new}} = 5 \div 2.23 = 2.24$$

- (c) By how much is the total execution time of the program improved if the CPI of Load instructions is reduced by 25%, and the CPI of branch instructions is reduced by 50%? **【Improvement = (Old Time - New Time)/Old Time】**.

$$(c) \quad \text{old time} = 1 \times 70 + 5 \times 25 + 4 \times 15 + 3 \times 30 = 345$$

$$\text{new time} = 1 \times 70 + 5 \times 25 \times \frac{3}{4} + 4 \times 15 + 3 \times 30 \times \frac{1}{2} = 268.75$$

$$\Rightarrow \text{Improvement} = \frac{345 - 268.75}{345} = 0.22 \#$$

6. (10%) Suppose you have a machine which executes a program consisting of 50% floating point multiply, 30% floating point divide, and the remaining 20% are from other instructions.

(a) Management wants the machine to run 4 times faster. You can make the divide run at most 3 times faster and the multiply run at most 8 times faster. Can you meet management's goal by making **only one** improvement, and which one?

$$(a) \quad \text{Case 1: } F = 30\% \quad S = 3 \quad \text{Case 2: } F = 50\% \quad S = 8$$

$$\frac{1}{0.7 + 0.1} = 1.25 \quad \frac{1}{0.5 + 0.0625} = 1.7777$$

(b) If you make **both** the multiply and divide improvements, what is the speedup of the improved machine relative to the original machine?

(b)

$$\frac{1}{\frac{0.5}{8} + \frac{0.3}{3} + 0.2} = \frac{1}{0.0625 + 0.1 + 0.2} = 2.758 \#$$

7. (15%) Consider the following two processors. P1 has a clock rate of 5 GHz, average CPI of 1, and requires the execution of 4.0×10^9 instructions. P2 has a clock rate of 4 GHz, an average CPI of 2, and requires the execution of 1.0×10^9 instructions.

(true/false 答對 2 分, clock rate 和 MIPS 大小對到 P1 < P2 的結果部分 3 分)

(a) Determine with the largest clock rate as having the largest performance. Check if this is true (or false) for P1 and P2, and why?

$$(a) \quad \text{Perf} = \frac{1}{\text{Time}} = \frac{\text{Clock Rate}}{\text{CPI} \times \text{Instruction}}$$

$$P1: \frac{5 \times 10^9}{1 \times 4 \times 10^9} = \frac{5}{4}$$

$$P2: \frac{4 \times 10^9}{2 \times 1 \times 10^9} = 2 \checkmark$$

A: No, in this case P2 with lower Rate have the higher Perf.

(b) Considering that processor P1 is executing a sequence of 1.5×10^9 instructions and that the CPI of processors P1 and P2 do not change, determine the number of instructions that P2 can execute in the same time that P1 needs to execute 1.5×10^9 instructions?

$$(b) \quad \text{Time}_{P1} : \frac{1.5 \times 10^9 \times 1}{5 \times 10^9} = \frac{1.5}{5}$$

$$\text{Time}_{P2} : \frac{\text{Instructions} \times 2}{4 \times 10^9} = \frac{1.5}{5}$$

$$6 \times 10^8 = 10 \times \text{Instruction}$$

$$A: 6 \times 10^8$$

(c) Calculate the processors P1, P2 MIPS (Millions of Instructions Per Second) separately, and determine that the processor with the largest MIPS has the largest performance. Check if this is true (or false) for P1 and P2, and why?

$$(c) \quad \text{MIPS}_{P1} = 5 \times 10^9 / 1 / 10^6 = 5 \times 10^3$$

$$\text{MIPS}_{P2} = 4 \times 10^9 / 2 / 10^6 = 2 \times 10^3$$

C/S	C/A
CLK	CPI

A: No, $\text{MIPS}_{P1} > \text{MIPS}_{P2}$, $\text{Perf}_1 < \text{Perf}_{P2}$

8. (10%) Translate the following C code to MIPS. Assume that the variables f, g, h, i, and j are assigned to registers \$s0, \$s1, \$s2, \$s3, and \$s4, respectively. Assume that the base address of the arrays A and B are in registers \$s6 and \$s7, respectively. Assume that the elements of the arrays A and B are 4-byte words:
 $A[h+5] = A[3] - B[g+1];$

VAR	<u>f</u>	<u>g</u>	<u>h</u>	<u>i</u>	<u>j</u>	<u>A</u>	<u>B</u>
REG	\$s0	\$s1	\$s2	\$s3	\$s4	\$s6	\$s7

```

lw    $t0, 12($s6)    # $t0 = A[3]
sll   $t1, $s1, 2      # $t1 = g * 4
addl  $t2, $t1, $s7    # $t2 = &B[g]
lw    $t3, 4($s7)      # $t3 = B[g+1]
sub   $t4, $t3, $t0     # $t4 = A[3] - B[g+1]
sll   $t5, $s2, 2      # $t5 = h * 4
addl  $t6, $t5, $s6    # $t6 = &A[h]
sw    $t4, 20($t6)

```

9. (10%) Translate the following MIPS code to C. Assume that the variables f, g, h, i, and j are assigned to registers \$s0, \$s1, \$s2, \$s3, and \$s4, respectively. Assume that the base address of the arrays A and B are in registers \$s6 and \$s7, respectively.

```

addi $t0, $s6, 16
lw   $t0, 0($t0)
sll  $t0, $t0, 3
sll  $t1, $s3, 2
add  $t1, $t1, $s7
lw   $t1, 0($t1)
add  $t0, $t0, $t1
add  $s4, $t0, $s1

```